1.4 Summary

- Many problems in the real world may be cast as expensive black-box optimization problems. In these problems, we only observe the function values without any additional information. Further, observing one function value is expensive, rendering many cost-blind optimization algorithms unusable.
- Bayesian optimization is a machine learning technique that solves this black-box optimization problem by designing intelligent evaluations of the objective function, so that the optimum may be found as quickly as possible.
- In Bayesian optimization, a Gaussian process acts as a predictive model, predicting what the value of the objective function is at a given location. A Gaussian process produces not only a mean prediction but also a 95% credible interval representing uncertainty via normal distributions.
- To optimize a black-box function, a Bayesian optimization policy iteratively makes decisions about where to evaluate the objective function. The policy does this by quantifying how helpful each data point is in terms of optimization.
- A Gaussian process and a policy go hand in hand in Bayesian optimization. The former is needed to make good predictions; the latter is needed to make good decisions.
- By making decisions in an adaptive manner, Bayesian optimization is better at optimization than random search or grid search, which are often used as the default strategies in black-box optimization problems.
- Bayesian optimization has seen significant success in hyperparameter tuning in machine learning and other scientific applications such as drug discovery.

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